**RAJARATA UNIVERSITY OF SRI LANKA**

**FACULTY OF APPLIED SCIENCES, MIHINTALE**

**B.Sc. (General) Degree**

# Tutorial 1 – ALGEBRA – MAP 2301

1. Let the operator $\Delta$ be defined by $A\Delta B = (A - B) \cup (B - A)$. Prove that,
   I. $A \cap (B\Delta C) = (A \cap B)\Delta(A \cap C)$.
   II. $A\Delta B = \emptyset$ if and only if A=B.
2. There are 100 students in a class and all the students are offered at least one of the three subjects chemistry, mathematics, and physics for an examination. 70 offered chemistry and 45 offered mathematics, and 45 offered mathematics. Further, 25 offered chemistry and mathematics, and 45 offered chemistry only or physics only.
   I. Find the number of students who offered chemistry or mathematics.
   II. How many students are offered physics only?
   III. How many students offered chemistry and physics but not mathematics?
3. a. Draw Venn diagrams to illustrate each of the following identities:
   I. $A - B = A \cap B'$
   II. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
   III. $(A \cap B)' = A' \cup B'$
   IV. $(A \cup B)' = A' \cap B'$
   V. $(A - B) \cap C = (A \cap C) - B$

   b. Given that $|A| = 55, |B| = 40, |C| = 80, |A \cap B| = 20, |A \cap B \cap C| = 17, |B \cap C| = 24$, and $|A \cup C| = 100$.

Find,

   I. $|A \cap C|$,
   II. $|(B \cup C) - (A \cap B \cap C)|$,
   III. $|(A \cup B \cup C)'|, if\ |\varepsilon| = 150$.
4. For any two subsets A and B of a given set S define A-B, the relative complement of B with respect to A. Using the definition of equality of two sets show that

I. $(A \cup B) - (A \cap B) = (A - B) \cup (B - A)$,
II. $(A - B) - C = A - (B \cup C) = (A - B) \cap (A - C)$,
III. $A - (B \cap C) = (A - B) \cup (A - C)$.

5. Let A, B abd C be subsets of a given set S.
   a. Prove that
      I. $If\ A \cap B = \emptyset\ then\ A \subseteq B'$,
      II. $A' - B' = B - A$,
      III. $A \subseteq B \Rightarrow A \cup (B - A) = B$,
      IV. $A \cap (B - C) = (A \cap B) - (A \cap C)$
   b. Let the operator $\Delta$ be defined by $A\ \Delta\ B = (A - B) \cup (B - A)$. Prove that,
      I. $A\Delta(B\Delta C) = (A\Delta B)\Delta C$
      II. $A \cap (B\Delta C) = (A \cap B)\Delta(A \cap C)$.
      III. $A\Delta B = \emptyset$ iff $A = B$.
6. Let A, B and C be subsets of a given set S. Using the Laws of Algebra of sets simplify the following expressions:
   I. $(A \cap B) \cup (B' \cap A) \cup (A' \cap B)'$
   II. $A \cup (A \cap B)' \cup [B' \cap (A' \cup C)] \cup (C \cap A')$
   III. $[A' \cap (B \cup C)] \cup [B' \cap (A \cup C) \cup [C' \cap B \cap A']$
7. Sixty students sat for an examination in mathematics, physics, and chemistry. 40 students passed in mathematics, 39 students passed in physics and 38 students passed in chemistry. If the number of students who passed only one subject is 6, shows that most 37 students passed all three subjects at least 23 students failed at least one subject.
8. It is observed that out of the teachers in a certain school, 44 are female, 54 are married and 34 are graduates. Further, 20 graduates are married and the number of male graduates exceeds the number of female graduates by 6.
   I. Show that the number of married female teachers in the school is the same as that of unmarried male graduates.
   II. Find the number of graduate female teachers in the school. Also, when two non-graduate unmarried male teachers in the school got married to two non-graduate unmarried female teachers in the same school, the number of unmarried male teachers became equal to the number of married female teachers
   III. Find the total number of teachers in the school.
9. Forty five students studying at the ‘CRIS’ International School offered Pure Mathematics, Applied Mathematics, Chemistry, and Physics respectively, but each student has passed at least one subject. Also, it is found that, among the students who passed physics, there were 12 who have failed in pure mathematics, 15 in applied mathematics, 13 in chemistry and 10 in exactly two subjects show that there were at least 5 students who have passed in physics only. Hiow any students have passed in chemistry only?
10. Of 100 rabbits it is found that57 are gray-coloured, 43 are male and 32 have long ears. 19 males and 21 gray-coloured ones have long ears, and there are 30 males who are gray.

Further, 20 of the rabbits are short-eared, non-gray-coloured females. How many gray-coloured, long-eared male rabbits are there?

11. Which of the following propositional forms are tautologies?
    I. $(p \vee q) \rightarrow (q \vee p)$
    II. $p \rightarrow ((p \vee q) \vee r)$
    III. $p \rightarrow (q \rightarrow (q \rightarrow p))$
    IV. $((p \rightarrow q) \leftrightarrow q) \rightarrow p$
    V. $(p \wedge q) \rightarrow (p \vee r)$
    VI. $(p \rightarrow q) \leftrightarrow (q \rightarrow p)$
12. Show by truth tables that $p \wedge (q \vee r)$ and $(p \wedge q) \vee r$ are not equivalent and that $(p \wedge (q \vee r)) \rightarrow (p \wedge q) \vee r$ is a tautology.
13. Prove, using truth tables.
    I. $p \rightarrow q \equiv (\neg q) \rightarrow (\neg p)$
    II. $\neg (p \rightarrow q) \equiv p \wedge (\neg q)$
    III. p$\rightarrow q \equiv (\neg p) \vee q$
    IV. $p \rightarrow q \equiv \left(p \wedge (\neg q)\right) \rightarrow O$
    V. Using part (iii), find a form equivalent to $p \leftrightarrow q$ which only uses the $\neg, \wedge, and\ \vee$ connectives.

       (It is because of parts (iii), (v) that the Boolean laws of logic can be written without → or ↔.)
14. Prove, using equation 7(iii) and other results from propositional logic, that $r \rightarrow$ (p $\rightarrow$ $q) \equiv (p \wedge r) \rightarrow q.$ so, do not use truth tables.
15. The "exclusive or" connective $\veebar$ is defined such that p$\veebar$q is true when either p or q are true but not both.
    I. Construct the truth table for p$\veebar$q,
    II. In words, we might say that "p$\veebar$q" is true if, and only if, "p$\vee$q" is true and it is not the case that p$\wedge$q is true". So we might think that p$\veebar$q$\equiv$ $(p \vee q) \wedge (\neg (p \wedge q))$. Use truth tables to show this is so.

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